



Principles Course Case Study Conclusion David Casaceli, PE

DALLAS, TX



DERAILMENT CASE STUDY

THE TWO L / V RATIOS



RILEY EDWARDS
UNIV. OF ILLINOIS U-C

1 TRACK STRUCTURES &
COMPONENTS

DARRELL KRUEGER
BNSF RAILWAY

2 VEHICLE TYPES, SUSPENSIONS,
AND COMPONENTS

KEVIN OLDKNOW
SIMON FRASER UNIV.

3 WHEEL-RAIL CONTACT: AN
INTRODUCTION

COREY PASTA
MxV RAIL

4 VEHICLE-TRACK INTERACTION &
DYNAMICS

RICHAR STOCK
PLASSER & THEURER

5 WHEEL-RAIL DAMAGE
MECHANISMS

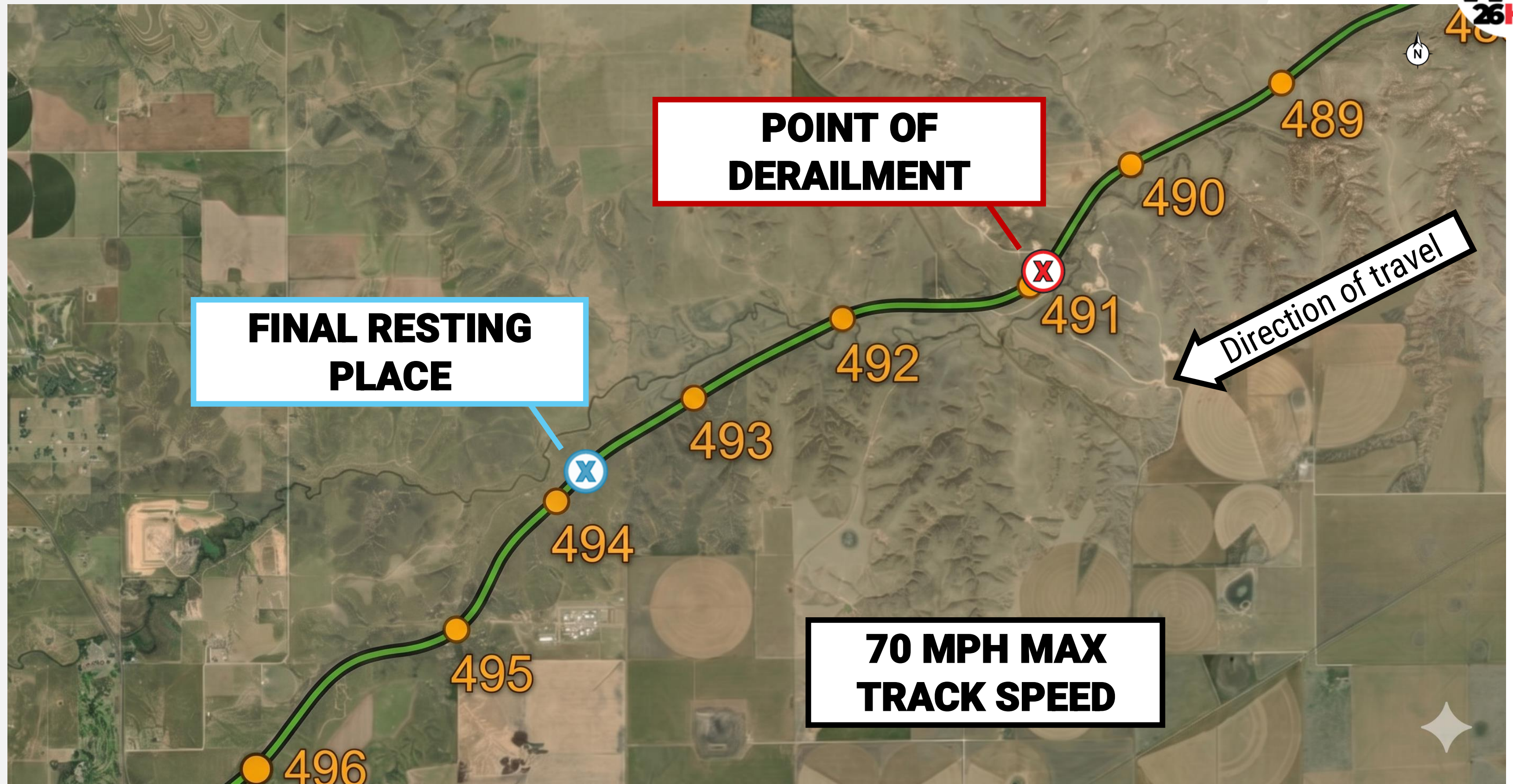
MATT DICK
ESi

6 VEHICLE-TRACK MEASUREMENT
TECHNOLOGIES

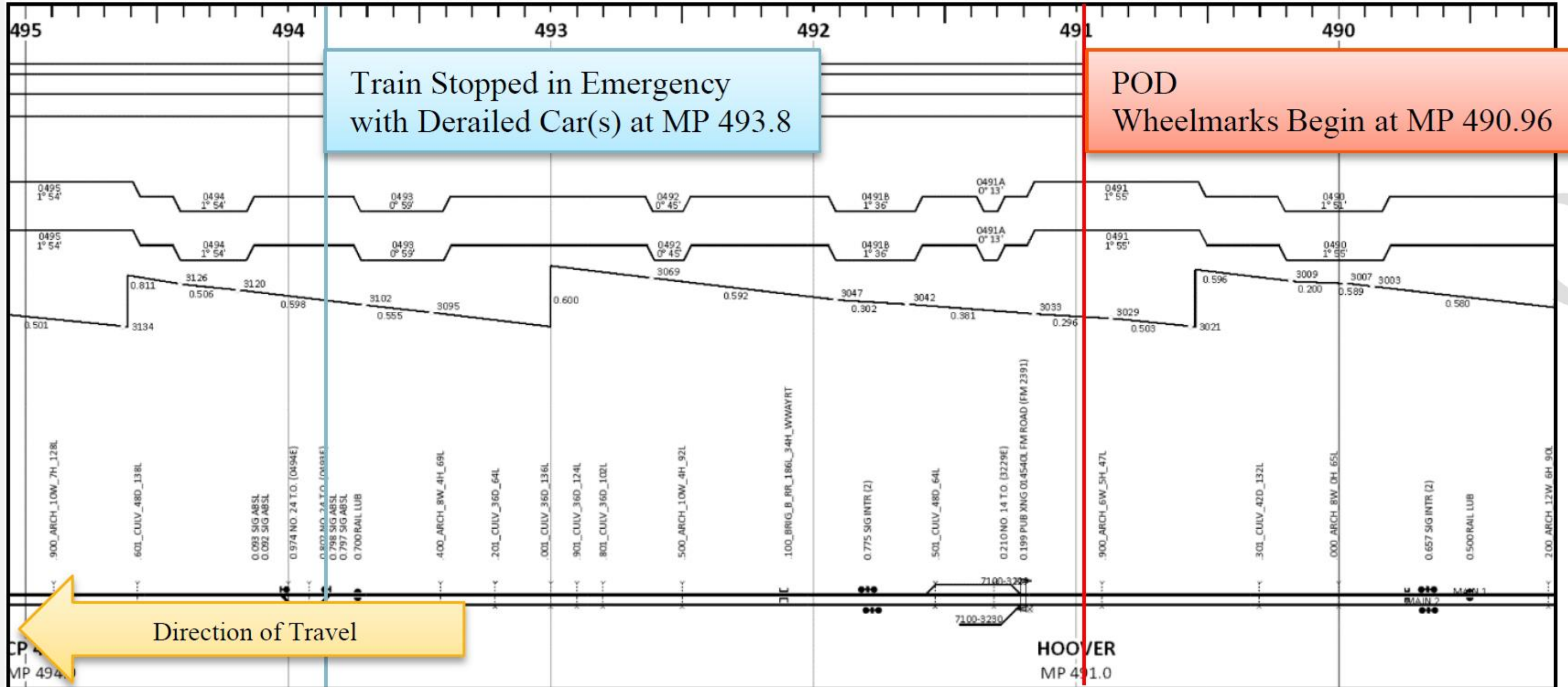








Source: FRA GIS MAP altered by Gemini

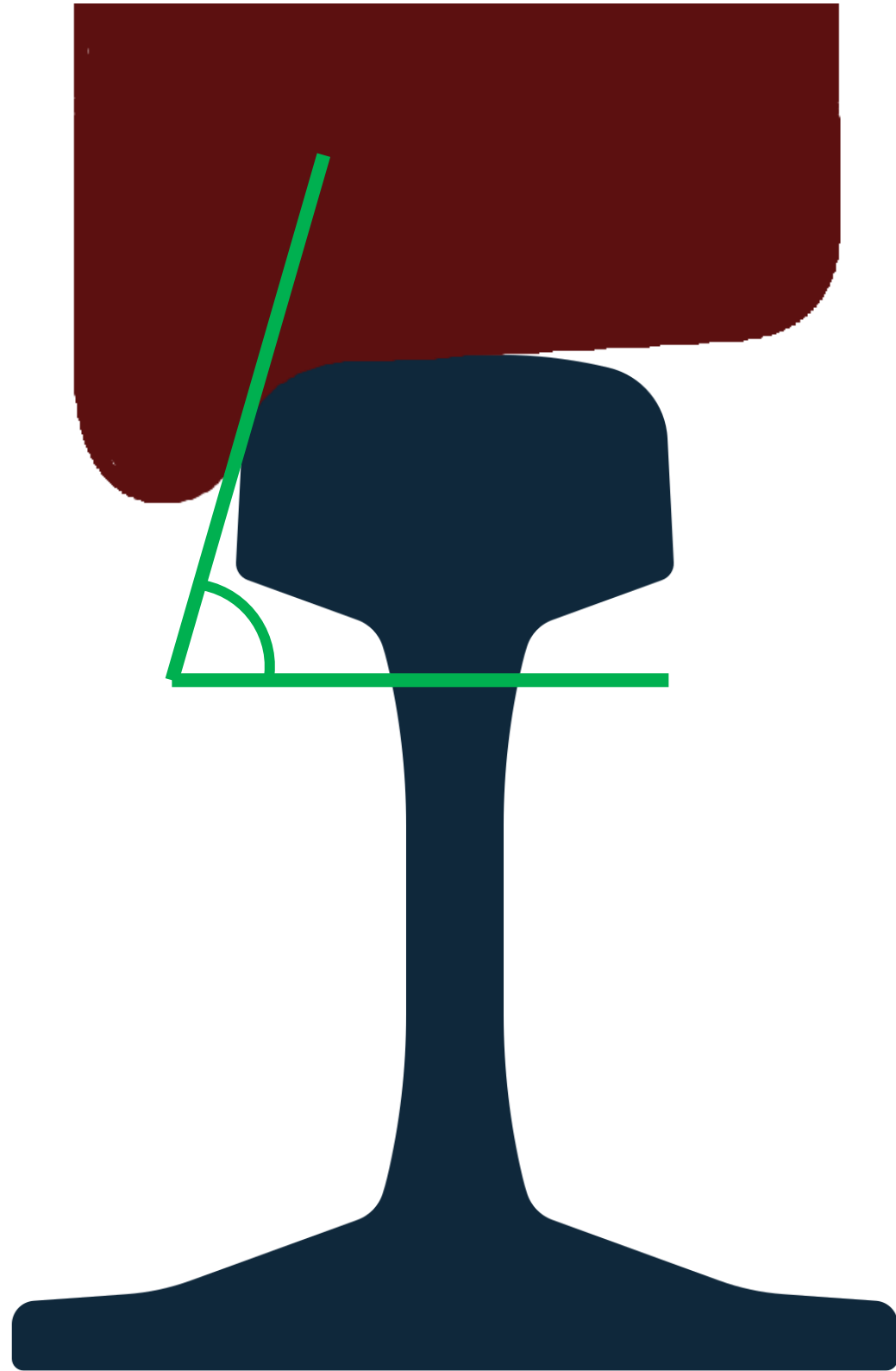




POINT OF DERAILMENT



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L/V THRESHOLD

WHEEL CLIMB

The threshold that, if exceeded , will result in a **wheel-climb** or rail-rollover for a specific wheel-rail combination.

L/V Threshold for WHEEL CLIMB is a function of:

- Angle at the interface
- Friction at the interface



APPLIED L/V

HOW APPLIED L/V CAN INCREASE?

BIG

$\frac{L}{V}$

LITTLE

V

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APPLIED L/V

WHEEL CLIMB

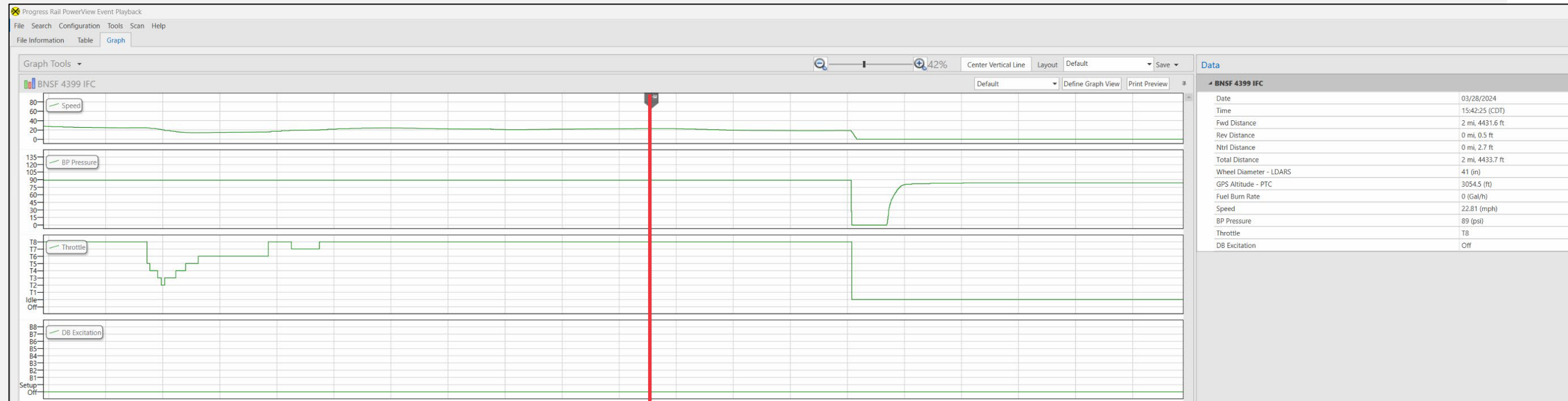
**MOST WHEEL CLIMB
DERAILMENTS ARE **LITTLE V****

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TRAIN MAKEUP & HANDLING

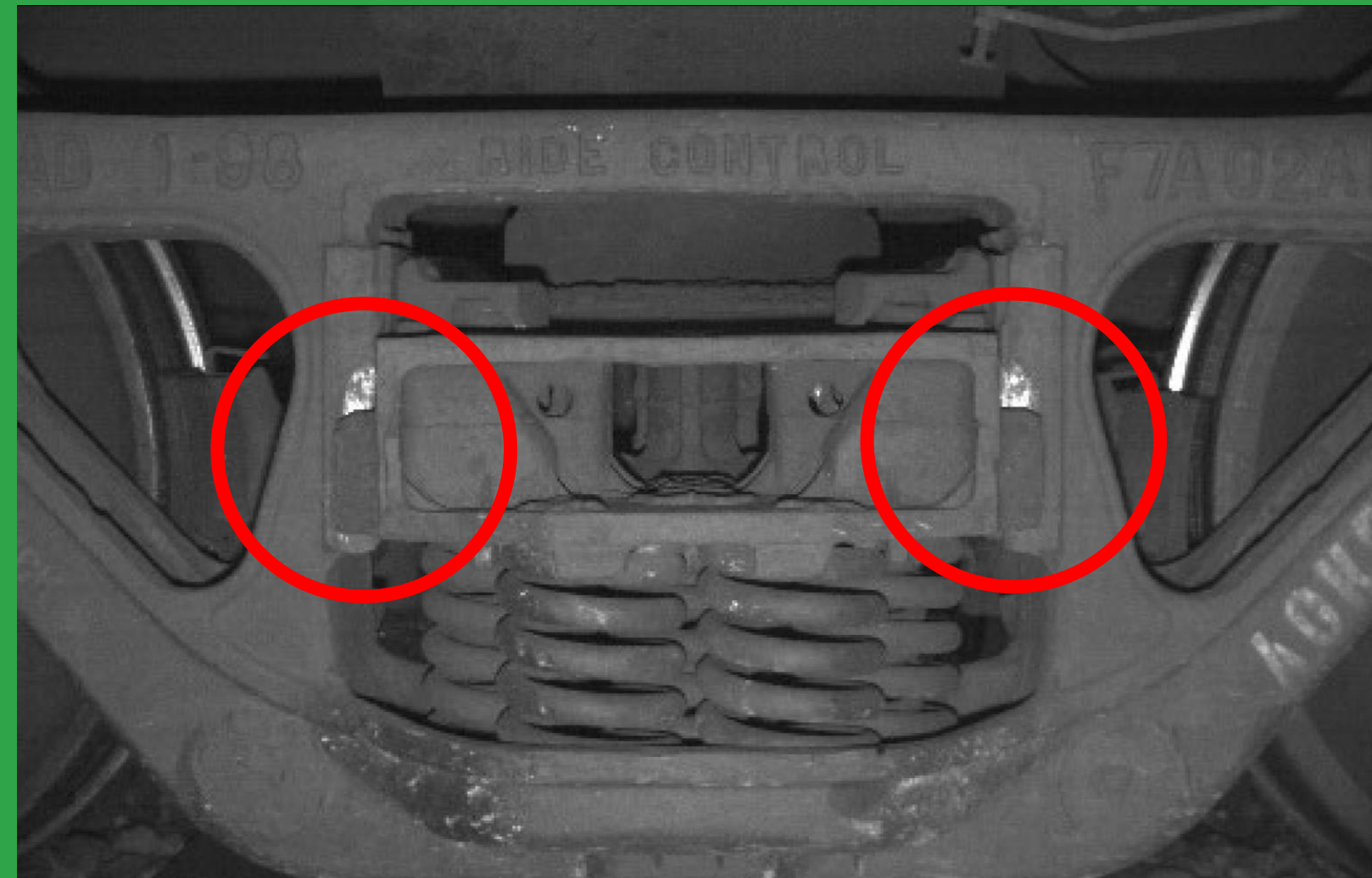


- ▶ 9,800 FEET & 7,200 TONS
3 LOCOS AT HEAD END
- ▶ SPEED = 23 MPH
- ▶ THROTTLE 8, STEADY
Uphill 0.4% Grade



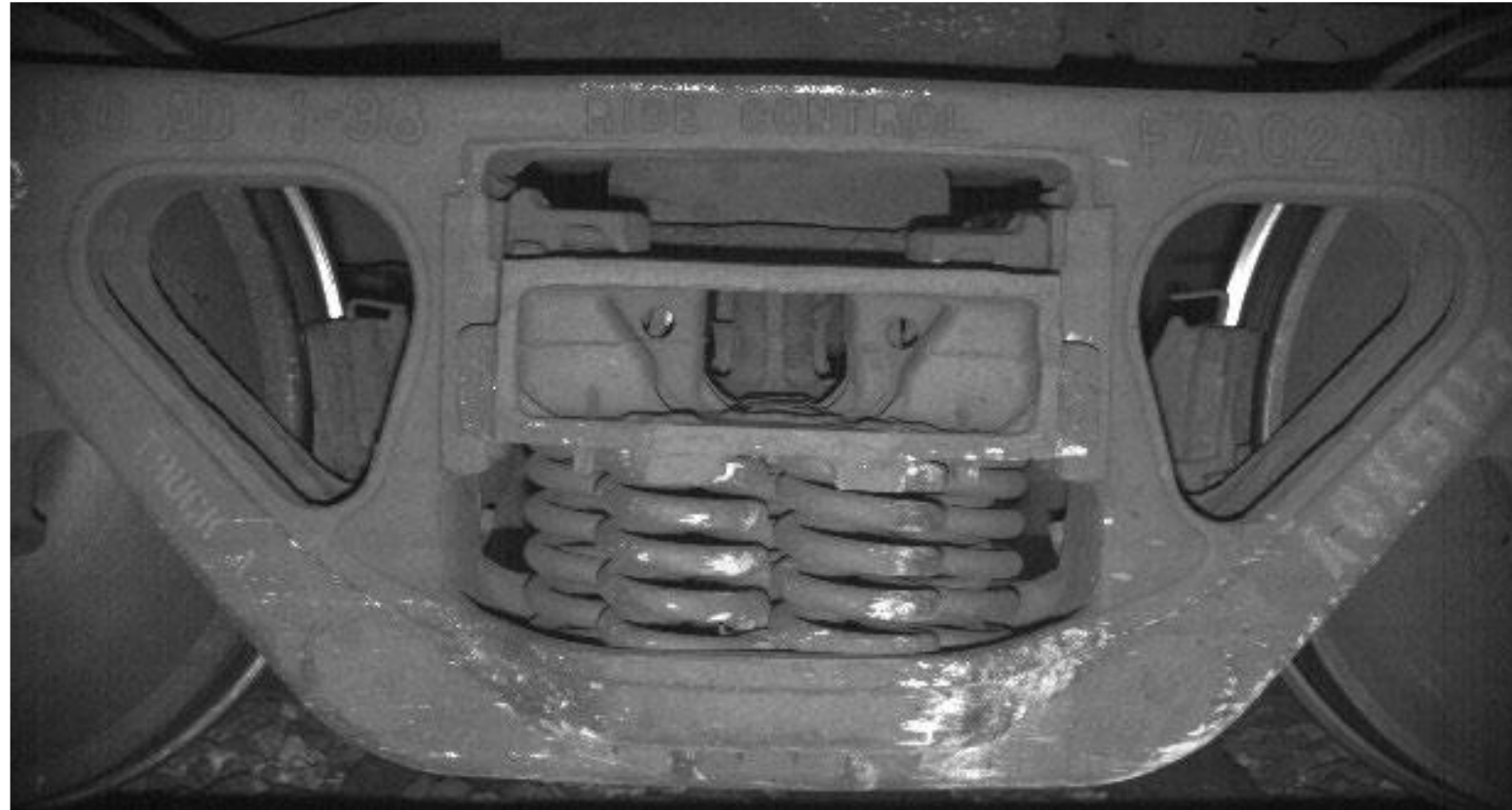
FIRST DERAILED CAR

- Determined by amount of tread damage to be AOK 5113
- 5-pack intermodal car
- Ride Control Trucks
- **WEDGE RISE**
 - Allowable = 1.82"
 - AR = 1.5" AL = 1.2"
- **GIB WEAR**
 - One side of A Truck





FIRST DERAILED CAR



- ▶ **ABRASION ON SIDE FRAMES & SPRINGS**
 - Appeared 6 – 8 weeks before accident
 - On left side of all trucks on AOK 5113
 - Car not previously reported in any derailment



FIRST DERAILED CAR LOADING

- ▶ **90.3" CENTER OF GRAVITY**
AAR HIGH CG = 98"
- ▶ **IMBALANCED LOAD TO THE RIGHT (LOW SIDE OF CURVE)**

$$\frac{L}{V}$$

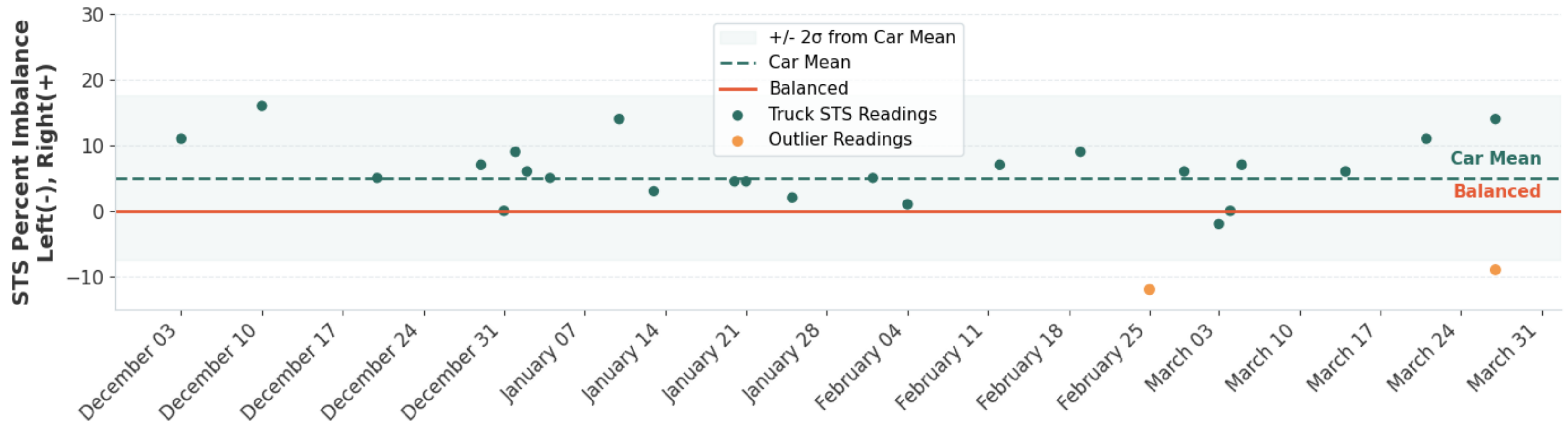
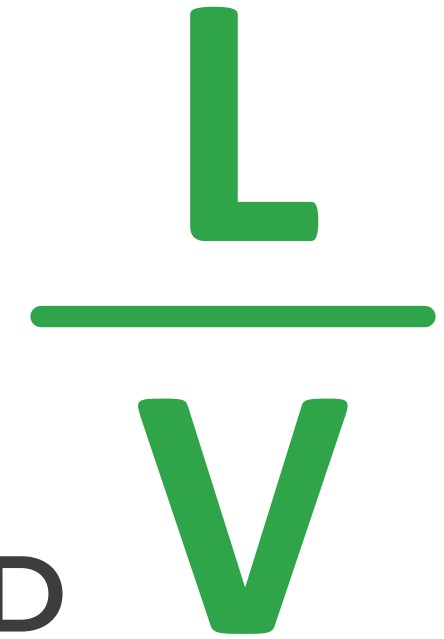
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FIRST DERAILED CAR IMBALANCE HISTORY

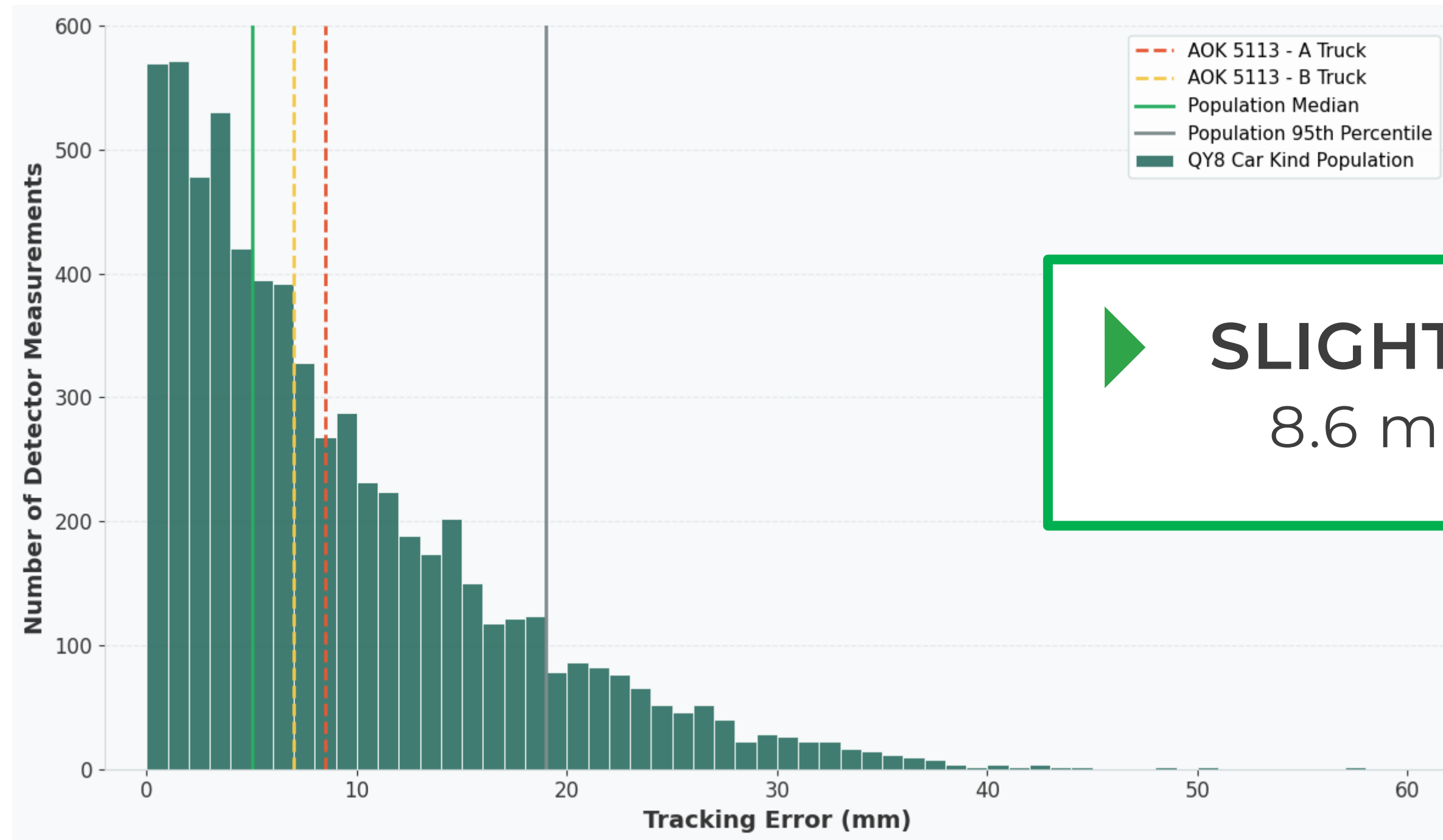
- ▶ WILD DATA SHOWS IMBALANCE HISTORY TO THE RIGHT – 6% Avg.
- ▶ DAY PRIOR: ACCIDENT TRAIN LEANED RIGHT 14% THEN LEFT 9%





FIRST DERAILED CAR

TRUCK GEOMETRY DETECTOR

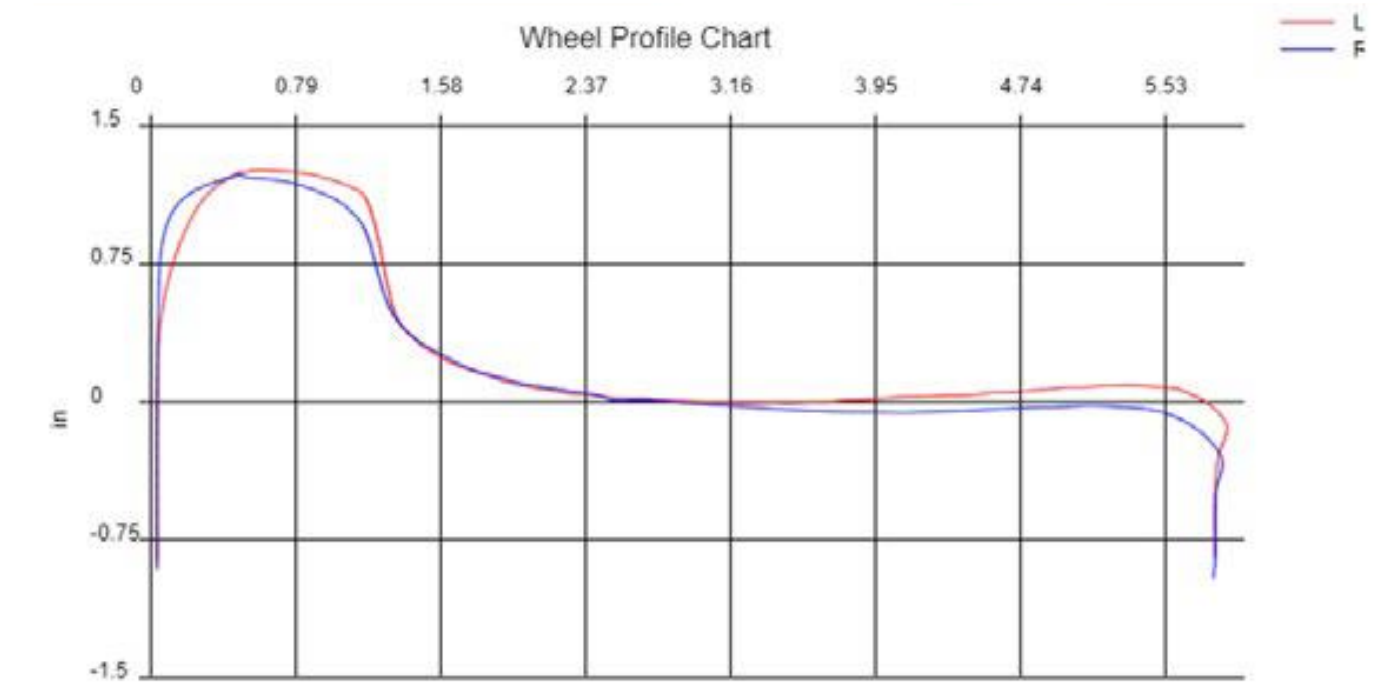


SLIGHT STEERING BIAS
8.6 mm (5 mm median)



FIRST DERAILED CAR WHEEL PROFILES

- ▶ FAIRLY SYMMETRIC WEAR
- ▶ SLIGHTLY HOLLOW on AL SIDE
2.5mm (4mm Allowed)





► **1.9 DEGREE RIGHT HANDCURVE**

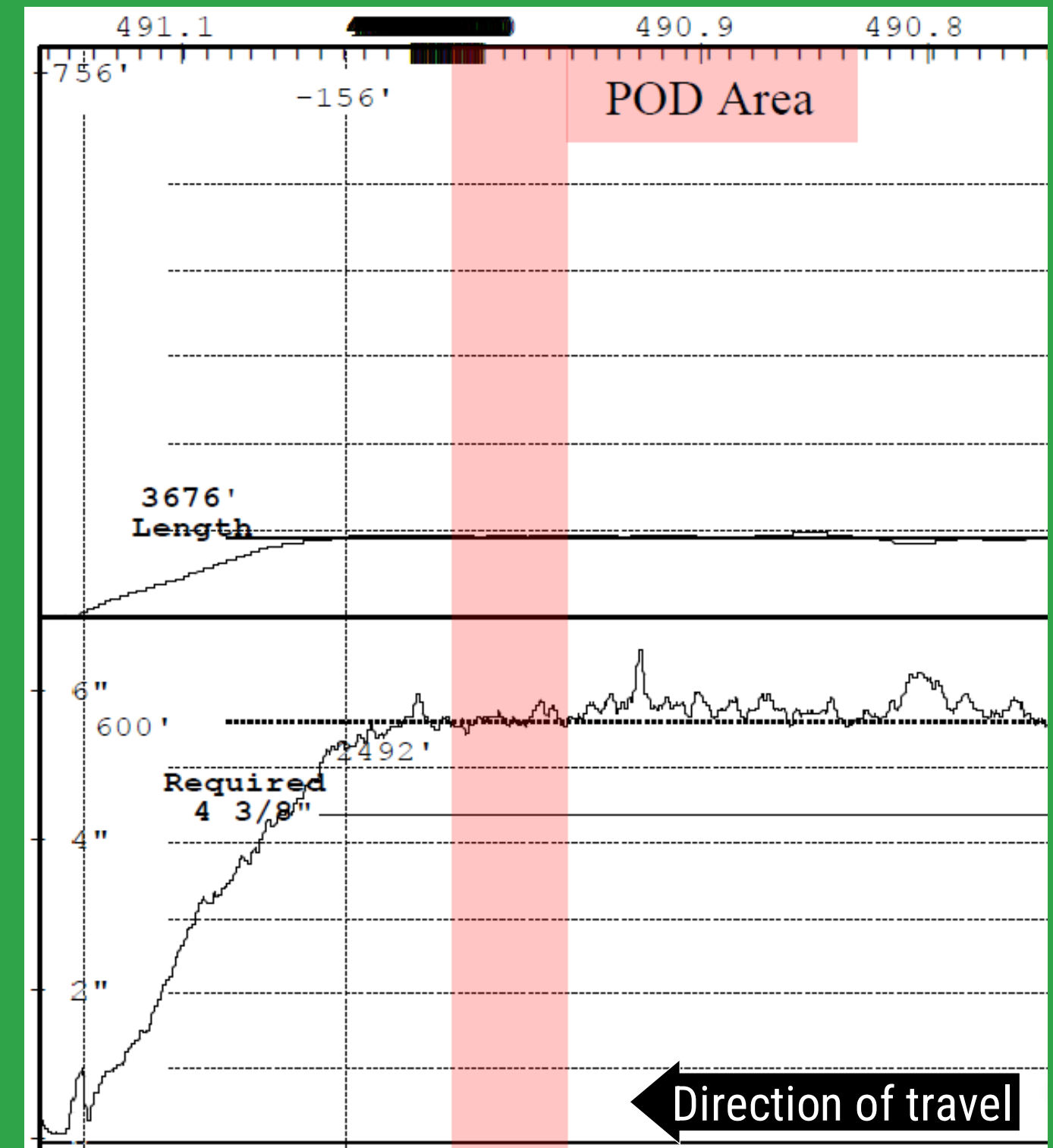
► **ABOUT 5-5/8" OF ELEVATION**
4-3/8" design

► **NO GEOMETRY DEFECTS**
Slight 1/2" Dip in low rail near POD
(FRA Allows 1 1/2" Warp)

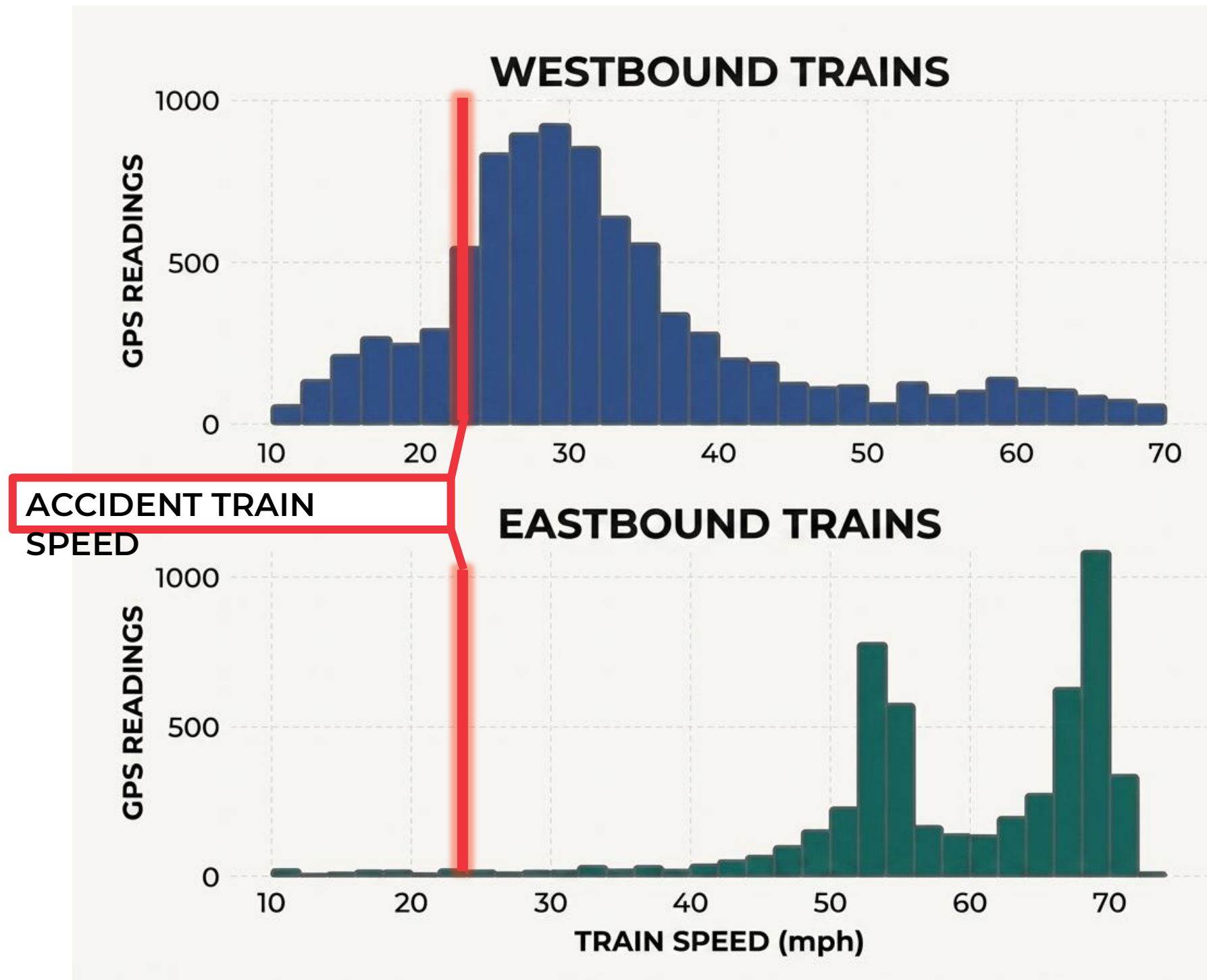
$$\frac{L}{V}$$

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TRACK & STRUCTURES



TRAIN SPEEDS & CURVE ELEVATION



Generated with Gemini

- ▶ **WHAT DO YOU SEE?**
Differential Speeds
- ▶ **WHAT DOES IT MEAN?**
Differing Balance
- ▶ **WHAT DOES IT MEAN FOR THE ACCIDENT TRAIN?**
Reduced Vertical Force on High Rail



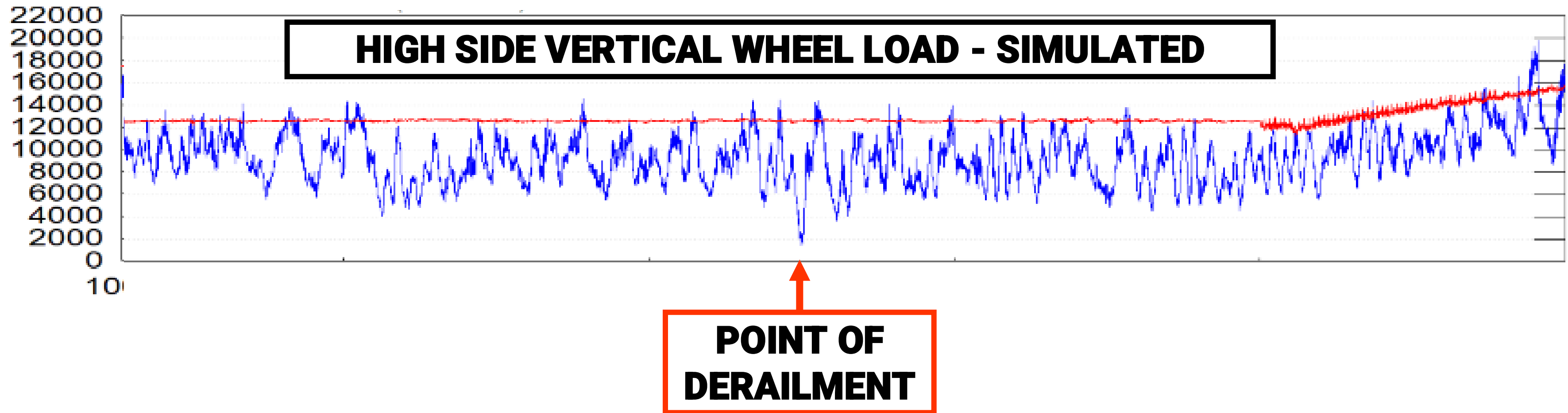


VEHICLE DYNAMIC SIMULATIONS





VEHICLE DYNAMIC SIMULATIONS



BLUE LINE = ACTUAL TRACK, CAR, AND LOAD CONDITIONS

RED LINE = NOMINAL TRACK, CAR AND LOAD CONDITIONS



VEHICLE DYNAMIC SIMULATIONS

Surface	Curve Elevation	Car	Load Shift	L1 Wheel Load (% of Static)
Nominal	Nominal	Nominal	Nominal	71%
Actual	Nominal	Nominal	Nominal	30%
Nominal	Actual	Nominal	Nominal	63%
Nominal	Nominal	Actual	Nominal	70%
Nominal	Nominal	Nominal	Actual	64%
Actual	Actual	Actual	Actual	10%

CASE STUDY CONCLUSION



▶ **Track Surface** and **Excess Elevation** had the greatest impact on vertical wheel load, but not sufficient to cause derailment.

▶ Elevated risk of wheel climb **required** the load shift **and** truck suspension/steering degradation.

▶ Derailment likely **required conditions not simulated. Such as:** Additional load shift, track surface degradation, or truck performance degradation.





CONTACT & RESOURCES



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